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Mapping and comparing climate equity policy practices using RAG LLM-based semantic analysis and recommendation systems

Seung Jun Choi¹ and Sangyup Lee^{2*}

Abstract

This study investigates the use of Large Language Models to enhance the policymaking process. We first analyze planning-related job postings to revisit the evolving roles of planners in the era of AI. We then present the Retrieval-Augmented Policy Analysis Framework (RAPAF), implemented using LangChain-native Retrieval-Augmented Generation pipelines with ChatGPT. The framework performs semantic analysis and extracts policy, strategy, and action items related to common themes in climate equity/justice or action plans. Based on these extracted elements and their evaluated presence, we develop a content-based recommendation system to support cross-city policy comparison. The results indicate that, despite growing attention to AI, planning jobs largely retain their traditional domain emphases in transportation, environmental planning, housing, and land use. Communicative responsibilities remain central to planning practice. Climate equity/justice or action plans commonly address transportation, environmental, and energy-related measures aimed at reducing greenhouse gas emissions and predominantly employ affirmative language. The implementation of our content-based recommendation system demonstrates how planners can efficiently identify cities with similar policy themes, highlighting patterns of cross-jurisdictional semantic similarity. The study concludes by envisioning localized and personalized AI-assisted systems designed to support professional planning judgment within urban governance contexts.

Keywords AI, LLM, RAG, LangChain, Recommendation system, Climate equity plan

1 Introduction

Since the emergence of Large Language Models (LLMs), including ChatGPT, Gemini, and Claude, and the subsequent development of agentic tools, recent years have seen growing efforts to deploy these systems to partially or fully substitute for human labor. In their simplest form, LLMs have been integrated into chatbot interfaces to support information retrieval and response generation (Ramjee et al., 2025). More recently, they have been synchronized with physical infrastructure, including autonomous vehicles (Chiu et al., 2025), traffic signals (Lai et al., 2025), and smart city systems (Kalyuzhnaya et al., 2025). Human labor associated with repetitive tasks, particularly

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work traditionally assigned to junior workers, is increasingly being replaced by AI-driven systems (Brynjolfsson et al., 2025), indicating that private firms have actively adopted these technologies in business operations.

In response to the ongoing paradigm shifts and increasing uncertainty in the field, this study investigates the extent to which AI can automate the extraction and comparative analysis of urban policy documents to support evidence-based planning. Specifically, it explores how AI-driven systems can streamline policy benchmarking by identifying semantic similarities and gaps across jurisdictions. Planning is an inherently interdisciplinary field and, epistemologically, has long advocated for the needs of disadvantaged groups through traditions such as advocacy planning and equity planning (Brooks, 2002). This orientation imposes an obligation on planners to advance the public good, typically operationalized through plans and policies. While planners must engage directly with communities and participate in field-based, natural language interactions (Sager, 2017), and while AI is likely to significantly reshape the planning workforce, it is unlikely to fully replace the core functions of planners.

Nevertheless, what if repetitive tasks could be replaced by AI, allowing human planners to increase productivity? Planners are often actively engaged in discussions that shape policy implications, yet they can only partially review planning documents because it is impractical to examine all relevant policies worldwide. It is also difficult to objectively determine whether a policy implemented in one city has been adopted, adapted, or omitted in another. The present study was developed in response to these challenges and seeks to utilize AI, specifically LLMs, for policy review. Related endeavors encompass the utilization of these methodologies for legal document classification (Wei et al., 2023) and the development of conceptual frameworks advocating their application in scenario planning (Kgomo & Song, 2025). Recent research, such as the work by Deng et al., (2025), has explored the use of ChatGPT to automate the collection of local extreme heat policies. However, their study was limited in geographic scope to Miami and Chicago, reviewed only eight documents, and did not employ accuracy-enhancing methods such as Retrieval-Augmented Generation (RAG).

This study explores how LLMs can be utilized for public policy analysis, with a particular focus on evaluating the presence and distribution of existing policies across multiple cities. In countries with large and complex governance structures, such as the U.S. and China, policy authority is distributed across multiple hierarchical levels, creating substantial challenges for systematic policy comparison. In the U.S., the interaction between home rule and federal and state authority results in layered planning documents, including comprehensive plans,

regional plans, and strategic plans. Given the central role of equity planning in contemporary planning practice (Brooks, 2002) and the growing emphasis on climate action to address climate change and advance sustainability goals (Aboagye & Sharifi, 2024; Franco et al., 2025), this study focuses on climate equity/justice or action plans in the U.S. In addition, the study revisits the professional obligations of planners by examining how planning work is evolving in response to AI adoption, drawing on a web-scraping analysis of planning-related job postings across the U.S. Examining planning job descriptions provides contextual insight into how AI-related analytical tasks may reshape the responsibilities of planners in practice. This analysis aims to revisit the roles of planners while demonstrating how AI can automate repetitive analytical tasks and contribute to a clearer conceptualization of planners' evolving responsibilities.

The objective of this study is to demonstrate how similar policy practices can be automatically matched and policy gaps identified through a recommendation system. This study focuses on how planning practice can be reformed, rather than on evaluating which AI model performs better. We envision a system in which public agencies, government bureaus, or individual practitioners can automatically collect policy documents and compare them based on similarity metrics, analogous to recommendation mechanisms used by platforms such as Netflix, YouTube, or search engines. The remainder of this article is structured as follows. Following the introduction, Sect. 2 reviews the relevant literature and provides the rationale for focusing on equity planning and the use of AI in planning practice. Section 3 describes the materials and methods. Section 4 presents the analytical results. Section 5 discusses the findings and their implications. Finally, Sect. 6 concludes the paper with a summary of the main contributions and directions for future research.

2 Literature review

2.1 Obligations of planners: equity and climate action planning

The origins of equity planning date back to the 1970s, when Norman Krumholz advanced the idea of redistributing resources and emphasized the role of planners as institutional activists. Since then, equity planning has become a major normative paradigm and a core professional obligation for planners (Brooks, 2002). Equity planning is not static; rather, it has continually evolved to address the dominant social and political challenges of each era. From a retrospective perspective, Zapata & Bates (2015) argue that equity planning has expanded across multiple fields, including health equity, climate planning, affordable housing, and planning pedagogy. They emphasize that equity planning is not confined

to formal planning bureaus. What matters most is not one's title, but the ability to build a collective vision and advance change across diverse institutional contexts, including community-based organizations, metropolitan planning organizations, and community activists (Zapata & Bates, 2015).

One of the urgent issues in contemporary planning is dealing with extreme climates. Climate extremes refer to infrequent weather or climate events identified relative to local historical climate conditions (Seneviratne et al., 2021). Ongoing climate change alters their likelihood and severity, increasing the occurrence of events that were previously rare or absent in the observational record (Seneviratne et al., 2021). Frequent exposure to natural hazards or irregular climate conditions is becoming the new normal. According to UN Environment (2022), recent climate observations indicate that temperature thresholds once considered exceptional are being exceeded with increasing regularity, reflecting a rapidly intensifying climate crisis.

Trenberth et al. (2015) argued that climate extremes should be assessed through their impacts, emphasizing how climate change amplifies damages given an event rather than focusing solely on causal attribution. This perspective introduces two complementary viewpoints. The first centers on environmental justice, emphasizing the need to address disproportionate impacts across communities. From this perspective, plan-making should be holistic, carefully considering both short- and long-term policy impacts while actively working to mitigate potential harms (Resnik, 2022). Meaningful participation by diverse community stakeholders and grassroots activists is essential, alongside recognition of procedural justice in decision-making processes (Schlosberg & Collins, 2014).

The second viewpoint emphasizes the necessity of structural change through concrete action. Karner et al. (2020) distinguished equity from justice, placing greater emphasis on the latter. Justice, in this framing, requires addressing fundamental structural barriers and unequal access to opportunities and is more closely aligned with bottom-up approaches to change (Karner et al., 2020). Within this context, climate action plans have emerged as policy instruments that translate climate concerns into actionable strategies for both mitigations, largely focused on reducing greenhouse gas emissions, and adaptation to evolving climate conditions (Stone et al., 2012).

Globally, climate action is closely connected to a wide range of development objectives, with implications extending across 16 of the UN's Sustainable Development Goals (Fuso Nerini et al., 2019). This body of evidence suggests that climate action should be conceptualized not merely as an environmental agenda, but as an integrated development strategy that is structurally linked

to poverty reduction, public health, urban sustainability, and institutional capacity building (Fuso Nerini et al., 2019). Comparative research further indicates that local climate action planning tends to concentrate on a limited number of core sectors. Across diverse national contexts, transportation and energy systems consistently emerge as the central domains of mitigation efforts, while other policy areas play more complementary roles. This sectoral pattern is evident in local planning frameworks in the U.S. (Deetjen et al., 2018) and Denmark (Damsø et al., 2016), as well as in national-level climate governance in China (Wu, 2023), where decarbonization strategies are largely oriented toward energy system transformation and transport-related emissions.

2.2 The use of AI in normative planning

Planning can broadly be categorized into positive and normative approaches. Normative planning engages in discussions of what planning practice *should* be (Brooks, 2002), drawing on multiple theoretical lenses to define what is considered equitable, justifiable, or sustainable. Equity planning represents one of the most prominent traditions within normative planning practice. Empirical analyses of equity concerns inherently involve multiple analytical frameworks. Such analyses often focus on identifying disproportionate burdens, including limited access to opportunities or heightened exposure to harms, experienced by specific socio-demographic groups (Karner & Niemeier, 2013). In transportation planning, equity analysis has largely centered on measuring accessibility inequality or accessibility poverty, approaches that are grounded in Rawlsian and sufficientarian philosophical traditions (Karner et al., 2025). These approaches are strongly influenced by earlier work in environmental planning, where scholars examined relative exposure to environmental harms, frequently operationalized through proximity-based measures (Bowen, 2002). Within this tradition, comparison groups are typically defined using frameworks derived from welfare and choice theory, spatial location theory, and empirical positivism (Bowen, 2002).

Normative planning practices are not rendered irrelevant in the era of AI; rather, they are being reconfigured and expanded through AI-enabled analytical and participatory capacities. Choi and Jiao (2024) argue that equity planning can evolve alongside AI by augmenting analytical practices while opening new pathways for community engagement. Their work demonstrates how AI-based transportation accessibility classifications can be integrated into interactive dashboards, enabling multiple stakeholders, including community members, to identify and interpret the spatial characteristics of their own locations. Beyond accessibility analysis, Generative AI (GenAI), including LLMs, has been shown to enhance

educational literacy (James & Andrews, 2024), transform user instructions into urban design content (Xu et al., 2024), and support further convergence in land use planning (Liu et al., 2025). Collectively, these applications align with Suh et al.'s (2023) argument that AI enables new forms of Human–AI (HAI) co-creation, fostering collaborative and iterative planning practices rather than replacing human judgment. At the same time, AI is already reshaping AI governance regimes in ways that diverge fundamentally from the smart city paradigm (Caprotti et al., 2024). LLMs are already widely used as analytical frameworks (Bai et al., 2026). Tiwari (2026) further extended this work by proposing a framework for developing Small Language Models (SLMs) to perform more localized urban informatics tasks. In this context, the key challenge for planners is not whether to adopt AI, but how to define thresholds and conditions for AI adoption that responsibly reconfigure contemporary planning practice while preserving its normative commitments with critical awareness (Seddone, 2025).

Additionally, recommendation systems are widely used in everyday contexts to support decision-making, such as purchasing products or selecting content on streaming platforms. As such, they constitute a critical component of personalized systems. Traditionally, platforms such as YouTube have relied on retrieval-and-ranking frameworks based on users' interaction histories, including viewing and liking behaviors, thereby establishing the foundational architecture for later deep learning-based recommendation systems (Davidson et al., 2010). More recent developments have extended these systems by embedding LLMs, enabling capabilities such as automated text description generation within recommendation engines (Acharya et al., 2023) and the simulation of user engagement patterns (Zhang et al., 2025). While recommendation systems have been extensively studied in commercial and media platforms, their application within the planning domain remains relatively underexplored.

Nevertheless, as urban systems become increasingly intelligent and data-driven, they are also likely to become more personalized, supporting both plan-making processes and everyday urban experiences. Building on this premise, the present study explores the potential of recommendation systems for policy matching across cities, illustrating how AI-driven personalization can be leveraged in urban and planning contexts.

2.3 Large language models and retrieval-augmented generation

LLMs have demonstrated strong capabilities across a wide range of natural language tasks. However, they still exhibit several limitations. One prominent issue is their tendency to hallucinate content, whereby models

generate factually incorrect or fabricated outputs that may appear superficially plausible (Huang et al., 2025). Additional constraints include a static knowledge cutoff and the inability to reliably attribute responses to specific sources, which can undermine reliability in high-stakes applications such as legal review, clinical decision support, and public policy analysis. RAG was introduced by Lewis et al., (2020) to address these limitations by combining pre-trained parametric language models with a non-parametric external retrieval memory. This architecture achieved state-of-the-art results on open-domain question-answering benchmarks and generated more factual and specific outputs than purely parametric models. A key advantage of RAG is that the document index can be updated without retraining the underlying model, enabling continuous incorporation of new knowledge in rapidly evolving domains such as climate policy.

LangChain was launched in October 2022 by Harrison Chase as an open-source framework for building applications with LLMs. It provides modular abstractions for document loading, chunking, embedding, vector storage, retrieval, and generation. LangChain integrates natively with dense embedding services and vector databases such as Facebook AI Similarity Search (FAISS; Douze et al., 2025), supporting scalable deployment across diverse institutional environments. The present study builds on these foundations by implementing a LangChain-native RAG pipeline for systematic extraction and cross-city comparison of climate equity/justice or action policies. Our pipeline employs overlapping chunking to preserve contextual continuity, FAISS-based dense retrieval, and Maximal Marginal Relevance (MMR) to balance relevance with diversity in retrieved passages (Carbonell & Goldstein, 1998). The temperature parameter is fixed at zero to ensure deterministic, reproducible outputs, following established practices in policy-sensitive and clinical applications (Ki et al., 2025; Kornblith et al., 2025). These design choices align with current best practices in RAG system design while addressing the specific characteristics of long-form, multi-city planning documents.

Recent literature increasingly advocates for a paradigm shift from static data analysis toward autonomous and multimodal urban intelligence systems through the integration of GeoAI, LLMs, and agentic systems for evidence-based urban planning (Huang et al., 2026; Li et al., 2024, 2026; Mai et al., 2024; Ye et al., 2026). GIS has long served as a foundational analytical tool in urban planning; however, GeoAI extends these capabilities by incorporating spatial autocorrelation, heterogeneity, and scale-sensitive analysis, thereby enabling the development of more autonomous and adaptive urban intelligence systems (Mai et al., 2024). Domain-specific studies further demonstrate this transition by highlighting the capacity of multimodal LLMs to interpret unstructured

spatial data and integrate heterogeneous urban datasets (Li et al., 2026). For instance, the BuildingMultiView framework illustrates how vision-language models can combine macro-scale satellite imagery with micro-scale street-view imagery to generate cross-scale architectural profiles, thereby overcoming the semantic fragmentation often present in conventional urban databases (Li et al., 2026). Similarly, StreetViewLLM demonstrates the effectiveness of integrating Chain-of-Thought (CoT) reasoning with RAG techniques to infer structured geographic insights from complex streetscape imagery and spatial coordinates (Li et al., 2024). Ki and Chen (2026) utilized ChatGPT with carefully designed prompts to evaluate walkability in Los Angeles, California, demonstrating that LLMs are capable of capturing nuanced microscale urban features beyond the object detection capabilities of conventional computer vision techniques. By decomposing complex spatial queries into sequential reasoning steps, these multimodal frameworks achieve greater granularity in capturing built environment characteristics and socio-spatial dynamics (Li et al., 2024). Collectively, these developments help bridge the semantic gap in human-centered urban analysis and support more context-aware urban intelligence frameworks (Mai et al., 2024; Ye et al., 2026). Building upon these emerging research directions, this study further explores how AI-assisted urban intelligence systems can support climate equity policy analysis and cross-city policy recommendation.

3 Materials & methods

3.1 Study area and materials

The study area includes U.S. cities with populations exceeding 100,000 that have adopted climate equity

plans. The list of cities was compiled from Wikipedia, which reports estimated population figures for 2024. Among the 346 cities considered, 192 (55%) had regional climate equity or action plans. The names of climate-related plans vary across cities; for example, New York City's PlaNYC: Getting Sustainability Done, Houston's Climate Action Plan: Resilient Houston, and Denver's 80×50 Climate Action Plan. We collected plans whenever climate-related themes were present and selected the most recent version available. The corresponding documents were collected from publicly available repositories in PDF format. Figure 1 illustrates the geographic distribution of cities with climate equity or action plans. While a small number of cities in Anchorage, Alaska, and Honolulu, Hawaii were identified, the majority of plans were concentrated in the contiguous U.S. For clarity of comparison and visualization, the analysis of policy practice differences focuses on the mainland U.S. Subsequently, these equity plans underwent a stepwise filtering process.

The present study also collected job descriptions to identify the responsibilities of planners in professional practice. The posted jobs were web-scraped on October 23, 2025, from LinkedIn and Indeed. We used the Python package JobSpy for data collection. Job postings containing the keywords communication, community engagement, community outreach, crime, environment, equity, finance, housing, planning, policy, public administration, public health, transportation, and urban were collected separately. They were selected based on planners' responsibilities and the presence of multiple sectoral planning themes (Brooks, 2002; Waddell, 2002). Using Natural Language Processing (NLP) techniques, irregular

Study Area: U.S. Cities with Climate Equity Plans

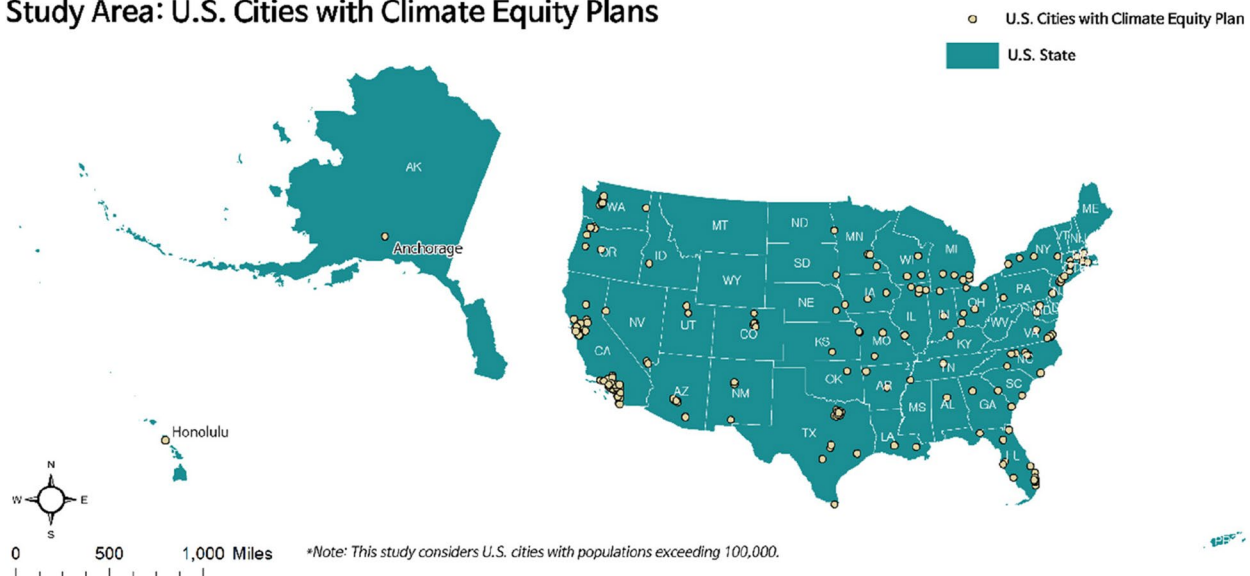


Fig. 1 Study area: U.S. cities with climate equity plans

characters, markdown syntax, bullet points, URLs, punctuation, random letters, extra spaces, hyphens, and dashes were removed. In total, 69,116 U.S. job postings were collected. After manually reviewing all entries, only 199 job postings remained.

Following established definitions of urban planning as a profession centered on public interest, spatial regulation, and long-term strategic development (APA, 2021; Fainstein & DeFilippis, 2015), we excluded postings where keywords were used in non-planning contexts. For instance, ‘equity’ often referred to financial stakes in private firms rather than social equity in urban zones; ‘community engagement’ frequently described classroom management for teachers or marketing for tourism; and ‘communication’ roles were often found in the hospitality sector rather than public participation in planning. These strict exclusion criteria ensured that the final dataset represented only those professionals directly responsible for the governance and physical planning of urban systems.

There were instances of duplicate listings, including multiple openings for the same position and repeated postings across different job announcements. After removing these duplicates, a total of 83 job postings remained. Figure 2 illustrates the workflow used to collect and filter job postings, including the number of postings identified for each keyword. While a substantial share originates from major metropolitan areas such as NYC, Los Angeles, San Jose, Houston, Miami, and Washington, DC, numerous postings are also located in mid-sized and smaller jurisdictions, including Lubbock (TX), Tacoma (WA), Edinburg (TX), Goldsboro (NC), and Bremerton (WA).

3.2 Methods

The methodology is structured in three interconnected stages to bridge professional workforce trends with AI-assisted policy analysis. First, to understand the evolving operational needs of practitioners, we analyze planning job postings. Driven by the findings that planners require tools to offload repetitive tasks and support decision-making, the second stage introduces the RAPAF pipeline to automate policy extraction. Finally, building on the structured outputs from RAPAF, the third stage develops a recommendation system to demonstrate how these extracted insights can support evidence-based intercity knowledge transfer and planning practice comparison.

Specifically, first, we conducted a text analysis of planning-related job titles and job descriptions. Word clouds were generated to visualize frequently occurring terms after removing stop words (e.g., and, or, for, the, by, of, with, etc.). The cleaned job descriptions were transformed into term–document matrices using term frequency–inverse document frequency (TF-IDF) weighting. Latent Semantic Analysis (LSA), implemented via truncated Singular Value Decomposition (SVD), was then applied to reduce dimensionality and extract latent topic structures. LSA-derived document–topic scores were used to examine thematic patterns across job descriptions, and the resulting topic intensities were visualized using a heatmap. The heatmap is used to visually examine the relative intensity and co-occurrence of LSA-derived topic scores across job descriptions, enabling comparison of thematic emphasis and identification of dominant and marginal roles in planning practice. For each topic, fifteen representative words were identified. The job-posting analysis establishes the operational need to automate repetitive document review.

Additionally, the same text-analytic approach was applied to review climate equity or action plans. We first compiled all terms appearing in the tables of contents across the collected climate equity or action plans. Generic section headings such as introduction, results, and appendix were treated as stopwords, as they do not convey substantive thematic meaning. The refined set of terms was then analyzed using the same text-analysis procedure to identify themes that are commonly addressed in climate equity or action plans.

Second, we introduce the Retrieval-Augmented Policy Analysis Framework (RAPAF) for the systematic review of climate equity policies. A key feature of RAPAF is the integration of a RAG mechanism to improve factual accuracy and contextual completeness compared with direct LLM prompting. Compared to direct LLM prompting, RAG grounds generation in retrieved source passages, substantially reducing hallucination risk and improving factual accuracy in knowledge-intensive tasks (Zhao et al., 2024). The pipeline was built using LangChain (Chase, 2022), which orchestrates the full indexing–retrieval–generation workflow through modular, composable abstractions.

This process consists of three steps: (1) indexing, (2) retrieval, and (3) generation. During the indexing stage, policy documents in PDF format were loaded and



Fig. 2 Flowchart of the job posting collection and filtering process

segmented using a recursive character-based text splitter with a chunk size of 1,000 tokens and an overlap of 200 tokens. The overlap window ensures the semantic continuity across segments. Each text chunk was embedded into a dense vector representation using OpenAI's text-embedding-3-small model, which maps text into a high-dimensional embedding space optimized for semantic similarity. The resulting embeddings were indexed in a FAISS vector store (Douze et al. 2025), which supports approximate nearest-neighbor search over large corpora in sub-linear time using inverted file structures and product quantization.

During the retrieval stage, each structured prompt was embedded using the same encoder at query time and used to query the FAISS index. Rather than standard top-k nearest-neighbor retrieval, we applied MMR (Carbonell & Goldstein, 1998), which iteratively selects passages that jointly maximize relevance to the query while minimizing redundancy among selected passages. MMR is defined as:

$$MMR = \operatorname{argmax} [\lambda \cdot \operatorname{Sim}(d_i, q) - (1 - \lambda) \cdot \max \operatorname{Sim}(d_i, d_j)]$$

where Sim denotes cosine similarity, q is the query embedding, d_i is a candidate passage, and d_j ranges over already-selected passages. We set $k=5$ (number of retrieved text chunks for generation), $\text{fetch_k}=20$ (initial candidate retrieval pool from FAISS), and $\lambda=0.7$ (weighting relevance over diversity). Random sampling was additionally applied over the candidate pool to further

mitigate retrieval bias toward topically similar but non-representative excerpts.

During the generation stage, retrieved passages were concatenated with a structured system prompt and passed to GPT-4o-mini as context. The prompt explicitly constrained the model to reason only from the provided passages and required citation of source documents and page numbers, creating an attribution chain that mitigates hallucination and supports auditability. The temperature parameter was fixed at 0, which collapses the output distribution to a deterministic argmax decoding, ensuring reproducible outputs across repeated runs (Peeperkorn et al., 2024). This practice is consistent with recent applications requiring deterministic LLM outputs in policy-sensitive and clinical contexts (Ki et al., 2025; Kornblith et al., 2025).

The RAPAF process using ChatGPT was adapted from Deng et al., (2025), who conceptualized policy content as a hierarchical structure consisting of policies, strategies, and actions. Figure 3 illustrates the workflow of policy extraction. Although document collection involved human oversight, we first applied an automated screening step to identify whether each document explicitly acknowledged challenges related to climate equity. ChatGPT was prompted to determine whether dedicated categories or sections addressing these challenges were present. When such sections were identified, ChatGPT was then instructed to extract relevant policies, strategies, and actions from the document. The extracted cases were compiled to identify prevailing contemporary practices. Our analysis focused on local practices related to

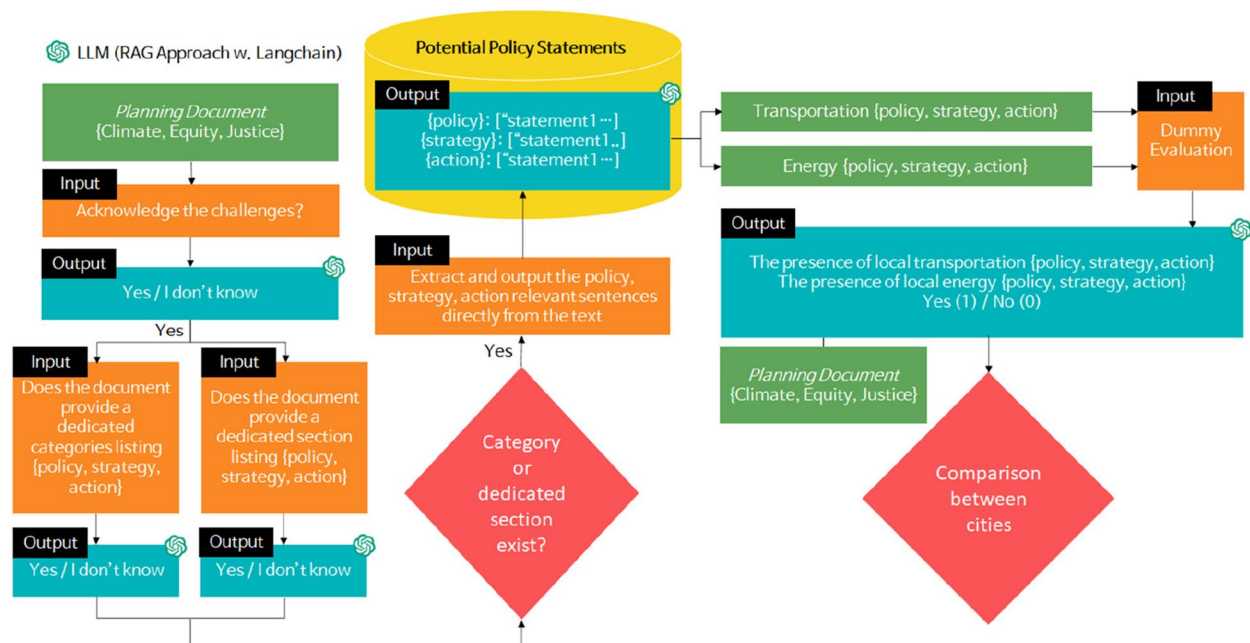


Fig. 3 Workflow of the RAPAF

Table 1 Identified contemporary themes in transportation policies, strategies, and actions from climate equity plans

Policy	Strategy	Action
Sustainable Transportation	Sustainable Transportation	Promote Bicycle Commuting
Transportation Equity	Transit System Improvement	Implement Carpool Programs
Multimodal Transportation Framework	Active Transportation	Expand Public Transit Services
Vehicle Emissions Standard	Low-Emission Vehicle	Reduce Vehicle Miles Traveled
Public Transit Accessibility	Transportation Demand Management	Encourage Active Transportation
Active Transportation	Bicycle Infrastructure Development	Develop Bicycle Lanes
Transportation Infrastructure Investment	Carpool and Vanpool Promotion	Enhance Transit Accessibility
Complete Streets Policy	Public Transit Expansion	Launch Transportation Awareness Campaigns
Bicycle and Pedestrian	Vehicle Idling Reduction	Install Electric Vehicle Chargers
Transportation Demand Management	Parking Management	Conduct Transportation Needs Assessments
Low-Emission Vehicle	Complete Streets Implementation	Implement Parking Management Solutions
Carpooling Incentive	Freight Efficiency	Support Vanpool Services
Transit-Oriented Development	Community Engagement in Transportation	Adopt Complete Streets Projects
Parking Management	Transportation Equity Strategies	Facilitate Community Workshops
Freight Transportation	Smart Mobility Strategies	Improve Transit Safety Measures
Transportation Safety	Integrated Transportation Planning	Encourage Use of Public Transit
Zero-Emission Vehicle	Green Transportation	Implement Traffic Calming Measures
Land Use and Transportation	Electric Vehicle Charging Infrastructure	Promote Shared Mobility Options
Transportation Resilience	Transportation System Optimization	Conduct Transportation Impact Studies
Smart Transportation	Land Use and Transportation Integration	Engage Stakeholders in Planning

transportation and energy, as these domains were most frequently addressed, as indicated by the table of contents analysis.¹ These policies generally aim to mitigate greenhouse gas emissions; for example, the transportation sector is the largest contributor to such emissions. This trend aligns with contemporary climate equity planning practices observed across diverse global contexts (Damsø et al., 2016; Deetjen et al., 2018; Wu, 2023).

Finally, the presence of each identified policy, strategy, and action was reassessed using structured binary prompts. This process was conducted across 20 predefined thematic categories. Table 1 and 2 present the list of thematic categories for transportation- and energy-related policy practices identified in contemporary climate equity plans. As illustrated, actions are generally more contextual and descriptive in nature compared to policies or strategies. These themes were identified through iterative semantic clustering of extracted policy, strategy, and action items generated by the RAPAF pipeline.² To ensure credibility, ChatGPT was instructed to

provide page numbers for all extracted statements and to respond with “I don’t know” when uncertainty arose.³ In total, the review process involved examining 21,489 pages comprising 5,834,348 words. The documents span the period from 2009 to 2025, with the majority of climate equity or action plans published between 2020 and 2025 (129 documents, representing 67%). The prompt designs are provided in Supplementary Table A.

The human-labeled ground truth was used as a reference standard to quantitatively evaluate the extraction performance of the LLM, providing a systematic validation of classification accuracy.⁴ The policy relevance extracted and evaluated by ChatGPT was compared against human-labeled ground truth across 192 climate equity and action plans. The overall average accuracy (overall proportion of correctly classified instance) was 70%, with relatively higher performance observed

¹ From the table of contents, excluding stop words, the words that appear at least 2 times were 523. After excluding adjectives and general terms such as plan, goals, and strategies, we identified the top ten themes in order of frequency. The most common theme was climate (189 occurrences). This theme was followed by terms related to greenhouse gas emissions, including greenhouse gas emissions (211 occurrences), emissions (77), greenhouse (50), gas (47), and GHG (37). Other frequently appearing themes included energy (81), community (70), transportation (66), sustainability (62, consisting of 44 occurrences of sustainability and 18 of sustainable), resilience (33), water (30), land (30), engagement (27), and so on.

² While Deng et al. (2025) manually constructed a list of local policy determinants, our primary objective was to explore pathways toward automating the policy recommendation system.

³ The identification of policies, strategies, and actions was based on semantic interpretation by an LLM rather than strict keyword detection. While this approach allows for greater flexibility in capturing substantively equivalent policy concepts expressed using different language, it may also introduce interpretive uncertainty. To mitigate this risk, the model was instructed to return page numbers and to respond with “I don’t know” when sufficient evidence was not present.

⁴ The human-labeled ground truth was generated through a structured manual coding procedure by a single expert coder certified by the American Institute of Certified Planners (AICP). For specialized policy documents, contextual expertise may contribute as much to analytical validity as inter-coder agreement (Bowen, 2009). The coding procedure was systematically conducted to construct a binary evaluation matrix (1/0) across the 192 climate equity/justice or action plans. Drawing upon practical experience in the planning profession, the expert was required to assign a positive label (‘1’) based on the presence of explicit formal headings and functional classification labels within the text.

Table 2 Identified contemporary themes in energy policies, strategies, and actions from climate equity plans

Policy	Strategy	Action
Clean Energy	Renewable Energy	Implement Energy Efficiency
Renewable Energy Standards	Energy Efficiency	Promote Renewable Energy Projects
Energy Efficiency	Decarbonization	Conduct Energy Audits
Energy Equity	Electrification	Launch Clean Energy Initiatives
Decarbonization	Energy Conservation	Support Electrification of Buildings
Electrification	Grid Modernization	Develop Community Solar Projects
Energy Conservation	Distributed Energy	Enhance Energy Storage Solutions
Sustainable Energy	Clean Energy Transition	Facilitate Energy Education Workshops
Grid Modernization	Carbon Reduction	Adopt Energy Conservation Measures
Distributed Energy Resources	Energy Storage	Engage in Carbon Reduction Activities
Clean Energy Access	Community Energy	Install Renewable Energy Systems
Fossil Fuel Reduction	Sustainable Energy Development	Conduct Public Awareness Campaigns
Energy Storage	Energy Access	Implement Smart Grid Technologies
Carbon Pricing	Smart Grid	Support Energy Transition Efforts
Energy Transition	Energy Resilience	Promote Sustainable Building Practices
Net-Zero Energy	Energy Innovation	Conduct Energy Impact Assessments
Energy Resilience	Fossil Fuel Phase-Out	Encourage Energy Innovation
Community Solar	Energy Demand Management	Facilitate Stakeholder Engagement
Energy Education and Awareness	Green Building	Implement Demand Response Program
Green Building Energy	Public Engagement in Energy	Support Local Energy Initiatives

in category-level classification compared to section-level classification. Specifically, section-level accuracy for policy, strategy, and action was 64%, 63%, and 68%, respectively. In contrast, category-level accuracy for policy, strategy, and action reached 64%, 79%, and 81%, respectively. The average precision (proportion of predicted positives that are true positives), recall (proportion of actual positives correctly identified), and F1-score (harmonic mean of precision and recall) across all tasks were 69%, 88%, and 77%, respectively. At the section level, policy classification achieved a precision of 55%, recall of 83%, and F1-score of 66%. Strategy classification showed a precision of 62%, recall of 97%, and F1-score of 76%, while action classification yielded a precision of 67%, recall of 98%, and an F1-score of 80%. At the category level, policy classification resulted in a precision, recall, and F1-score of 65% each. Strategy classification achieved a precision of 83%, recall of 91%, and F1-score of 87%, whereas action classification showed a precision of 83%, recall of 94%, and F1-score of 88%. Overall, the results indicate that the model tends to achieve high recall across most tasks. However, comparatively lower precision in certain cases, particularly at the section level. Figure 4 illustrates the confusion matrices comparing LLM predictions with human-labeled ground truth, highlighting true positives, true negatives, false positives, and false negatives.

While traditional frequency-based models (e.g., TF-IDF, LDA) are useful for identifying broad thematic clusters, they inherently rely on bag-of-words assumptions that dismantle syntactic structure and fail to capture

deep contextual nuances (Egger & Yu, 2022). Consequently, conventional models output isolated tokens that are structurally inadequate for distinguishing hierarchical policy intents, such as differentiating a general policy statement from a specific, actionable measure (Williams et al., 2024). Therefore, to systematically extract the nuanced policy–strategy–action hierarchy and perform deterministic semantic classification, this study fundamentally necessitated the advanced reasoning capabilities of an LLM-based RAG framework.

A sample of text excerpts and their corresponding evaluations by the LLM, compared with human-coded ground truth, are presented in Table 3. The sample is drawn from the New York City climate action plan, which represents the most populous city in the U.S. According to the human planner's evaluation, the plan explicitly and comprehensively covers a wide range of sections and categories. When identifying explicit section headings, the LLM closely matched the human-coded ground truth. However, in category identification, several cases revealed that human evaluators were better able to recognize nuanced semantic intentions and contextual meanings than the LLM, resulting in noticeable differences between the two evaluations.

The performance of LLM-based extraction is generally moderate and varies substantially depending on task complexity and prompt design, indicating that while LLMs can be effective, they are not consistently reliable across contexts. Prior studies report a wide range of evaluation outcomes, highlighting the sensitivity of performance to both task characteristics and evaluation metrics

Confusion Matrices of LLM Predictions versus Human-Labeled Ground Truth

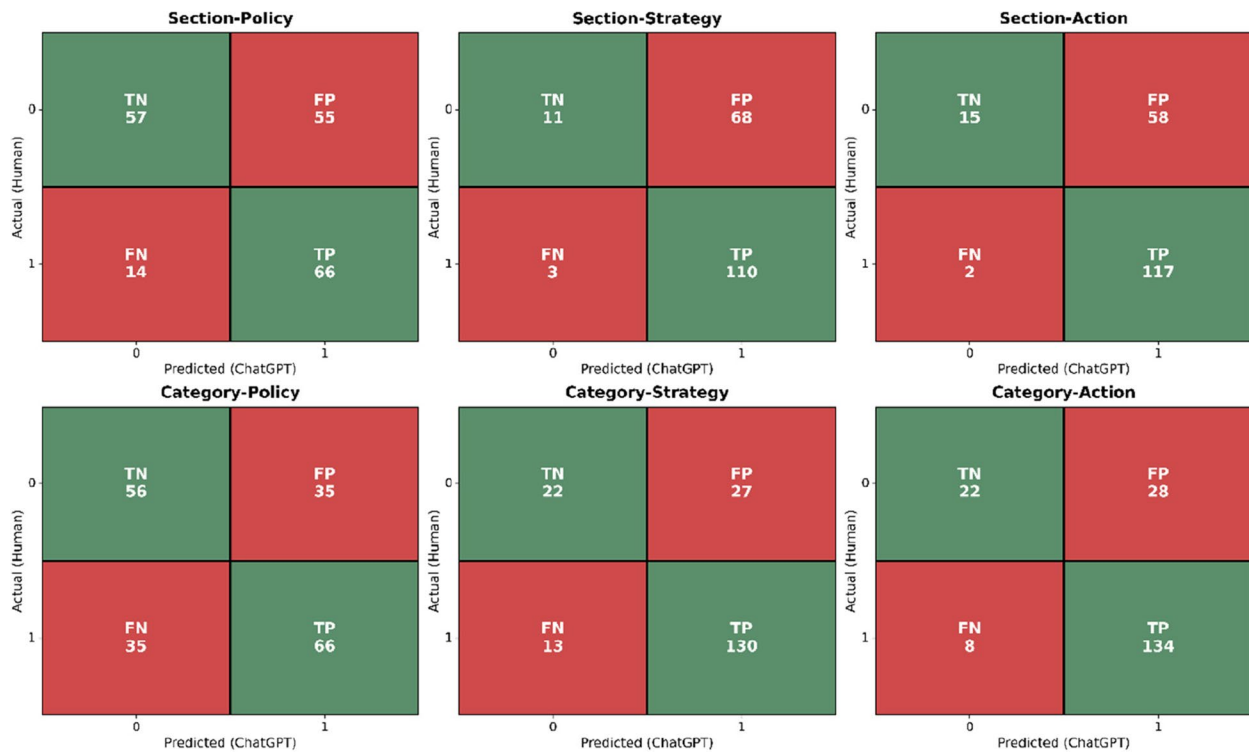


Fig. 4 Confusion matrices of LLM predictions versus human-labeled ground truth

(Adam et al., 2025; Stewart-Evans et al., 2025; Yu et al., 2026). For instance, in structured clinical review and classification tasks, LLM-based extraction has achieved high accuracy levels of 93–98% (Adam et al., 2025). However, in other scenarios such as medical document review and annotation, performance has been substantially lower, with F1-scores ranging from 10 to 55% (Yu et al., 2026). Similarly, in policy-oriented scoping reviews, LLMs have demonstrated relatively high accuracy (approximately 80%) and precision (over 90%), but considerably lower recall and F1-scores, often below 25% and 40%, respectively (Stewart-Evans et al., 2025).

Third, policy practice comparisons were conducted using a recommendation system framework. A content-based approach was employed, in which similarity was derived from the attributes of policy content itself rather than from observed interactions or diffusion patterns. Paired matrices representing cities and their associated policies, strategies, and actions were constructed. Cosine similarity was then used to identify the top five most similar city pairs. For each policy element, the rate of adoption among similar cities was calculated as the mean of binary indicators. Based on these similarity scores, we evaluated whether specific policies, strategies,

and actions were commonly practiced or whether gaps existed across cities. Representative examples were presented to illustrate the results.

To facilitate interpretation and practical use of the recommendation results, we implemented an interactive search-based interface that allows users to query a target city and retrieve comparable peer cities along with recommended policy actions. Through this interface, users can input or select a city name and dynamically view the most similar cities identified by the cosine similarity metric, as well as policy, strategy, and action elements that are commonly adopted by peer cities but absent in the target city's plan. This search-based implementation enables transparent exploration of policy similarities and gaps across cities and supports reproducible inspection of the recommendation outcomes (Deng et al., 2025; Gupta et al., 2022). The interface was designed as an analytical aid rather than a decision-making tool, providing an interpretable mechanism to examine how current climate equity or action policies vary across jurisdictions. Policy-makers and public agencies would benefit from similar interfaces and tools to efficiently support policy narrative drafting. The interactive interface was implemented using ipywidgets within a virtual computing environment.

Table 3 Comparison of human-coded and LLM-extracted results: a case study of New York City's PlaNYC (2023)

Category		Text snippet from document	LLM extracted	Human coded (Ground Truth)
Section	Policy	<i>referenced by LLM</i> • Climate and environmental policies (IRA, LL97); climate governance and environmental justice (MOCEJ); green economic development and clean energy transition (NYCEDC); building decarbonization and energy efficiency; sustainability, equity, and climate action initiatives	1 (Present)	1 (Present)
	Strategy	<i>referenced by LLM</i> • Multilayered climate adaptation strategy; neighborhood-scale coastal protection; property-scale flood resilience measures; flood resilience standards development; shoreline asset protection; sea-level rise and storm adaptation planning	1 (Present)	1 (Present)
	Action	<i>referenced by LLM</i> • Carbon neutrality implementation actions; renewable energy expansion; City of Yes for Carbon Neutrality initiative; citywide zoning and text amendments; implementation milestones; Department of City Planning (DCP) led climate actions	1 (Present)	1 (Present)
Category	Policy	<i>referenced by human</i> • The plan has a dedicated section named “Protecting Us from Climate Threats” and list dedicates categories of policy (e.g., maximizing climate infrastructure on city-owned property; prioritizing public transit, etc.)	0 (Absent)	1 (Present)
	Strategy	<i>referenced by human</i> • Similar to policies organized under distinct categories, the plan also includes dedicated strategies addressing climate threats related to extreme heat, flooding, buildings, and clean and reliable energy	0 (Absent)	1 (Present)
	Action	<i>referenced by LLM</i> • The document does not explicitly mention categories for actions such as grouping by theme, sector, priority, timeframe, or implementation phase. It primarily discusses various initiatives and milestones without categorizing actions in the specified ways. Therefore, I don't know <i>referenced by human</i> • The climate action plan comprehensively outlines action items across a range of climate-related themes, including climate hazards, green spaces, waterways, transportation, and food systems	0 (Absent)	1 (Present)

4 Results

4.1 Textual and thematic analysis of planning and policy narratives

Figure 5 illustrates the results of topic modeling based on text analysis of job descriptions and job titles extracted from job postings. The heatmap presents topic modeling scores derived from job descriptions, highlighting the relative prominence of each topic. Across job descriptions, the most frequently occurring terms were planning (876), experience (499), development (404), work (375), city (310), public (301), transportation (296), related (276), projects (242), information (219), and environmental (216), among others. Job titles predominantly included keywords directly related to planning roles, such as planner (54) and planning (29). Other commonly appearing terms included transportation (17), GIS (11), urban (11), director (10), assistant (9), analyst (9), environmental (7), city (5), and operations.

Based on the LSA-derived topic scores, Topic 1 was primarily centered on city (0.149), transportation (0.149), and development (0.141), reflecting general urban and transportation planning roles. Topic 2 was strongly associated with transportation (0.242), the Metropolitan Transportation Authority (MTA) (0.135), administration (0.105), and management (0.096), suggesting alignment with managerial or director-level positions (0.088).

Topic 3 also emphasized transportation (0.220) but highlighted technical and regulatory expertise, including zoning (0.166), GIS (0.154), and software proficiency such as ArcGIS (0.124), along with knowledge of permits (0.119) and land use (0.104). Topic 4 was most strongly associated with intensive use of GIS (0.236) and a technical orientation (0.142), with an emphasis on data analysis (0.134). This topic also showed close associations with housing (0.136) and real estate (0.108). Topic 5 similarly reflected strong GIS usage (0.266), combined with expertise in transportation (0.210), environmental (0.156), regional (0.148), and housing (0.134) domains, indicating a multidisciplinary planning profile.

Figure 6 further illustrates the topic modeling and text analysis based on both the table of contents and the RAG-LLM-generated responses to the prompt: “What are the principal challenges and pressing issues addressed in this regional plan regarding the promotion of social equity and community advocacy?” Analysis of the planning documents directly extracted from the table of contents shows that climate (195), plan (118), and action (112) were the most frequently occurring keywords. Other prominent terms included energy (81), emissions (77), community (73), transportation (66), reduction (59), greenhouse (50), gas (47), waste (46), and sustainability (44). In comparison, the RAG-LLM-generated responses

Text Analysis of Planning Job Descriptions and Titles

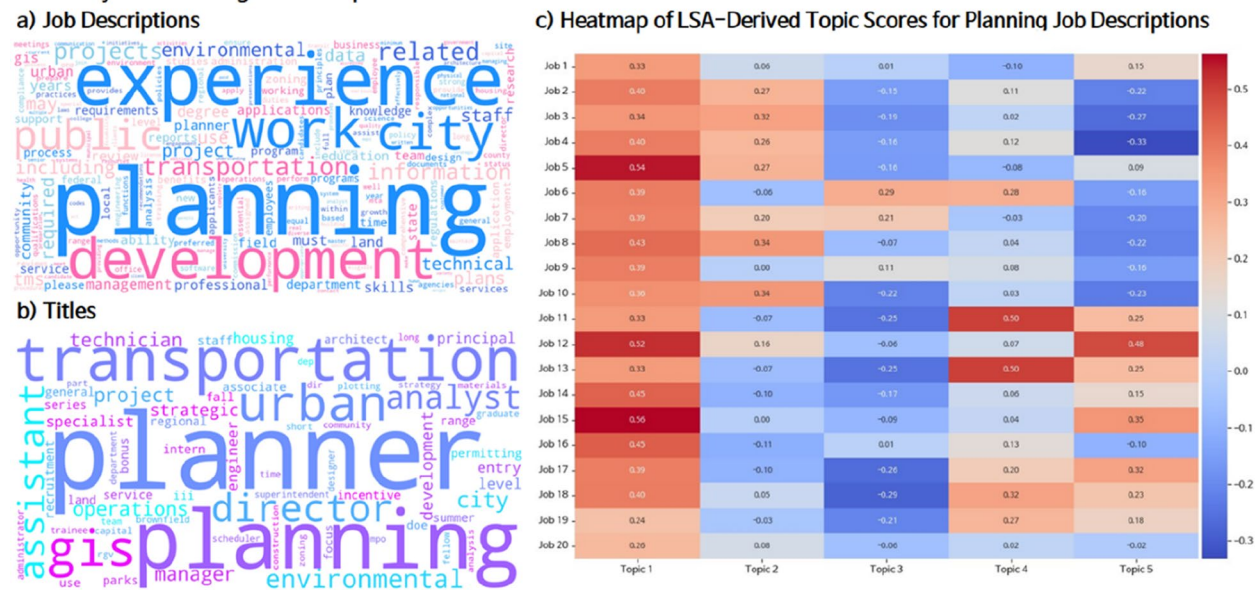


Fig. 5 Text analysis of planning job descriptions and titles: **a** Word cloud of job descriptions; **b** Word cloud of job titles; **c** Heatmap of LSA-derived topic scores across planning job descriptions

Text Analysis of Climate Equity Plans

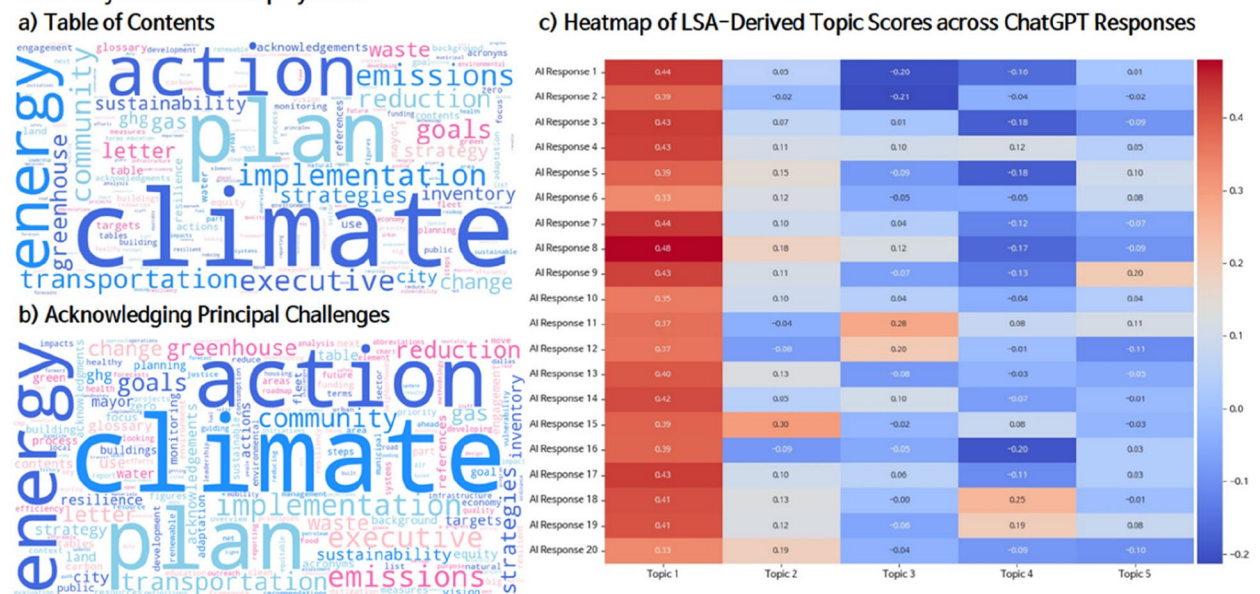


Fig. 6 Text analysis of climate equity plans: **a** Table of contents; **b** Acknowledging principal challenges; **c** Heatmap of LSA-derived topic scores across ChatGPT responses

exhibited a stronger emphasis on equity-oriented language, with frequent occurrences of climate (171), communities (131), access (107), equitable (89), engagement (88), residents (76), income (67), health (65), change (59), vulnerable (58), action (58), populations (58), necessity (58), housing (55), transportation (53), low (52), environmental (51), strategies (50), enhance (50), and energy (50).

Each sentence in the generated responses was scored along the topic dimensions, and the top-ranked sentences for each topic were selected as representative statements, capturing the core semantic content of each topic in a concise and interpretable manner. Topics 1–5 represent distinct yet interconnected dimensions of climate equity discourse across the analyzed plans. Topic 1 reflects a broad climate action framing that emphasizes communities (0.171), access (0.138), equitable

(0.120), and engagement (0.117). Topic 2 centers on food security, with food (0.192), access (0.119), and services (0.104) emerging as dominant terms. Topic 3 relates to the promotion of sustainability and long-term environmental considerations. Topic 4 is primarily associated with energy-related themes, showing the highest score for energy (0.313), while also incorporating systematic (0.113), historical (0.127), and inequities (0.123). Topic 5 is largely centered on food (0.500), with additional links to housing (0.139), health (0.112), and infrastructure (0.107).

Table 4 presents representative phrases derived from topic scores, demonstrating that climate equity/justice-oriented plans span multiple planning domains, including environmental policy, food security, housing, transportation, energy, and infrastructure. The LLM successfully identifies underlying structural needs by explicitly acknowledging disadvantaged groups, such as low-income populations (e.g., LIDAC) and communities of color (e.g., BIPOC). Moreover, action-oriented policy items reflecting existing local conditions are consistently extracted, highlighting the applicability of the approach for diagnosing equity-focused planning priorities.

4.2 Semantic review of climate equity plans using LLMs

To further assess the linguistic characteristics of the extracted responses, a transformer-based language model was employed to derive semantic polarity scores. Specifically, a BERT-based classifier fine-tuned for binary sentiment classification was applied to the LLM-generated responses corresponding to policy sections, strategy

sections, and action sections, as well as their respective categorizations. For each response, the model produced both a categorical label (positive or negative) and a confidence score. These outputs were transformed into signed polarity values to preserve directional information, allowing negative polarity to indicate statements emphasizing absence, ambiguity, or lack of explicit mention, and positive polarity to reflect affirming or articulated content. The semantic polarity analysis was not intended to evaluate policy effectiveness but to diagnose how explicitly or assertively climate or equity measures were articulated across planning documents.

In most cases, existing climate equity or action plans predominantly use affirmative language when listing policies, strategies, and action items. Among documents that did not include “I don’t know” responses, negative classifications were identified in 19 policy-related phrases, 19 strategy-related phrases, and 17 action-related phrases at the section level. When aggregated by category, the corresponding counts were 17, 27, and 31, respectively.

Tables 5 and 6 present representative samples classified as positive and negative cases with the highest confidence scores. These samples are derived from section-level phrases. The positive cases include relevant sentences from Albuquerque, Sacramento, and Cleveland, while the negative cases are drawn from Brownsville, South Bend, and Tampa. Affirmative cases typically include language that supports future directions and planned actions. In contrast, negative cases often reflect policy language that acknowledges limitations, challenges, or unresolved issues. For example, a phrase from Brownsville explicitly

Table 4 Common principal challenges in promoting equity and advocacy in practice

Topic	RAG-LLM responses generated using LangChain
Topic 1 Climate Access & Equity Promotion (keywords: climate, communities, access, importance, equitable, engagement, promotion, action)	<i>The principal challenges and pressing issues addressed in the regional plan regarding the promotion of social equity and community advocacy include the need for direct engagement with LIDAC (Low-Income and Disadvantaged Communities) populations, ensuring that climate action initiatives are inclusive and supportive of these communities</i>
Topic 2 Community Well-Being & Sustainable Services (keywords: communities, food, residents, access, sustainable, services)	<i>The principal challenges and pressing issues addressed in the regional plan regarding the promotion of social equity and community advocacy include the disproportionate impacts of climate change on marginalized and vulnerable populations, such as low-income communities and BIPOC (Black, Indigenous, and People of Color) individuals</i>
Topic 3 Climate Sustainability and Policy Implementation (keywords: sustainability, climate, promotion, change, quality)	<i>The principal challenges and pressing issues addressed in the Westminster Sustainability Plan regarding the promotion of social equity and community advocacy include the need for an inclusive approach that ensures diverse groups have equal rights, opportunities, and access to community capital programs and services</i>
Topic 4 Energy Transition and Environmental Justice (e.g., energy, food, climate, efficiency, disparities, inequities, emissions, healthy, systemic)	<i>The regional plan addresses several principal challenges and pressing issues related to social equity and community advocacy, including the need for enhanced community engagement in climate action initiatives, the promotion of energy efficiency and renewable energy access for all residents, and the reduction of automobile emissions through improved public transit and pedestrian infrastructure</i>
Topic 5 Food Systems, Housing, and Essential Services (e.g., food, climate, local, housing, production, health, action, infrastructure)	<i>The regional plan addresses several principal challenges and pressing issues related to social equity and community advocacy, including the need to overcome barriers to food access in underserved communities, such as "food deserts," by encouraging the development of grocery stores and expanding food assistance benefits like SNAP/EBT/WIC (Supplemental Nutrition Assistance Program/Electronic Benefit Transfer/Special Supplemental Nutrition Program for Women, Infants, and Children)</i>

Table 5 Representative samples classified as positive with the highest confidence scores

Items	RAG-LLM responses generated using LangChain
Policy	from Albuquerque, NM • Yes, the document includes a dedicated section listing policies. The relevant sentences are: "Support local and state legislation that prioritizes urban infill, brownfield redevelopment and renovations, rather than new developments/new construction." and "Support consistent and timely adoption of local and state legislation that requires developers and home builders to continue to meet current energy standards for newly constructed or renovated buildings and homes."
Strategy	from Sacramento, CA • Yes, the document provides a dedicated section listing strategies. It mentions several key City initiatives that address sustainability and equity, including the establishment of the City's Office of Diversity & Equity, the formation of the City Council's Racial Equity Committee, the establishment of the City's Office of Climate Action & Sustainability and the Climate Work Plan, and the development of an Inclusive Economic Development Strategy and Action Plan
Action	from Cleveland, OH • Yes, the document provides a dedicated section listing actions. It states that "Actions are clear, specific, actionable steps that the City, its partners, and its residents can take to complete objectives and put our community on the path to meeting our goals." Additionally, it mentions that "the City of Cleveland has analyzed each action using the Evaluative Framework laid out in the previous chapter" and identifies "a handful of Priority Actions within each Focus Area that the City will focus on implementing immediately." Furthermore, it highlights a set of Cross-Cutting Priorities essential for advancing climate justice, which include Circular Economy, Empowerment, Equity, Good, Green Jobs, and Resilience

Table 6 Representative samples classified as negative with the highest confidence scores

Items	RAG-LLM responses generated using LangChain
Policy	from Brownsville, TX • Yes, the document includes a dedicated section listing policies. The relevant sentences are: "No public money incentives will be paid to companies that do not pay living wages. Living wage clause will be included in municipal purchase contracts. All contracts awarded by the municipality (or by public entities in Brownsville) would require contractors and all sub-contractors to pay living wages to employees. For new business, bidding credits will be awarded to bidders that pay living wages. A bidder that pays living wages would get a credit in their submitted bid equal to a specified percentage of the submitted bid." This is found in the section titled "Strategic Initiatives" on page 7
Strategy	from South Bend, IN • Yes, the document includes a section on strategies. The relevant sentences are: "While useful climate strategies can be pulled from around the globe, this plan prioritizes solutions that most clearly fit South Bend's institutions, demographics, natural environment, regional economy, and infrastructure. Oftentimes, the highest-impact strategies are not the most feasible. Considering the importance that this plan be actionable for the city, strategies were prioritized based on how achievable they were for the community, municipality, and other agencies, and whether they proved to be too expensive or at the cost of other priorities, like quality of life, access, and economic opportunity." This information is found in the section titled "INVENTORY AND METHODOLOGY" on page 6
Action	from Tampa, FL • Yes, the document includes a dedicated section listing actions under the "List of Projects, Programs, Policy and Administrative Changes" on page 146. However, the specific actions are not detailed in the provided context

limits the potential impact of the policy. The case of South Bend similarly highlights explicit cost constraints. There are instances in which a dedicated section exists but lacks substantive detail, as illustrated by the case of Tampa, Florida.

4.3 Identification of similarities and gaps in climate equity plans and practices

In the evaluation of transportation and energy policy, strategy, and action themes within climate equity or action plans, no cases achieved a full score of 20 out of 20 for the respective criteria. On average, transportation themes scored 7 for policy, 6 for strategy, and 8 for action across the evaluated cities. Land use and transportation integration emerged as the most common transportation policy theme, with transportation resilience receiving the least attention. For transportation strategies, electric vehicle charging infrastructure was the most common theme, whereas smart mobility and integrated

transportation planning were the least represented. For transportation actions, the installation of additional electric vehicle charging infrastructure was most frequently identified, while launching transportation awareness campaigns, conducting needs assessments, and conducting impact studies were the least commonly observed actions.

Across the energy themes, the average scores were 9 for policy, 8 for strategy, and 6 for action across the evaluated cities. Energy efficiency was the most frequently addressed theme in energy policy, while grid modernization was the least addressed. Renewable energy was the most common focus for energy strategies, while energy innovation and grid modernization received the least attention. Among action items, implementing energy efficiency measures was most common; however, conducting formal energy impact assessments was observed in only one case.

Based on the evaluation results, we constructed a content-based recommendation system. Recommendations are generated based on the similarity of evaluation patterns across action-related indicators. It is important to note that this similarity does not reflect geographic proximity between cities; rather, it indicates similarity in the action patterns recorded in the evaluation data. Because action items are more detailed and context-specific, we demonstrate a matching exercise in which cities with high scores are compared to others with similar practices. Figure 7 illustrates the retrieval of transportation action items for the case of Las Vegas. Las Vegas exhibited analogous transportation action patterns to those of Chico, Ann Arbor, Berkeley, Richmond, and Long Beach. Across these cities, common practices include promoting shared mobility options, facilitating community workshops, and launching transportation awareness campaigns. However, the current documentation appears to lack action items explicitly focused on enhancing transit accessibility, supporting vanpool services, and conducting needs assessments. Similarly, Fig. 8 demonstrates the search for comparable cases for energy-related action items using Fort Lauderdale as the reference city. Fort Lauderdale shares similarities in practice with Saint Paul, Des Moines, Sterling Heights, Columbia, and Buffalo. Commonly observed actions include supporting energy transition efforts and facilitating stakeholder engagement. In contrast, actions such as launching clean energy initiatives, offering energy education workshops, adopting energy conservation measures, and conducting public awareness campaigns are likely missing.

5 Discussion

This study demonstrates how AI-based text analysis and recommendation systems can support the comparison, diagnosis, and learning of planning practices for climate equity. While AI does not replace planners' normative judgment, the findings show that such systems can function as analytical tools that reveal patterns embedded in large volumes of policy documents and expand the range of policy options available to practitioners. The analytical findings are as follows.

First, in the era of AI, this study revisits the evolving role of planners and their professional responsibilities. Explicit references to AI were largely absent from the analyzed job postings, despite frequent requirements for GIS and data analysis skills. Domain-specific planning areas such as transportation, environmental planning, land use, and housing remained central across positions. These roles were not confined to traditional planning job categories but extended across sectors, including business, management, finance, education, social services, technology, intelligent transportation systems, and logistics, and were observed in both public and private organizations. Taken together, the topic modeling results suggest that planning jobs are evolving toward a hybrid professional model that combines coordination-oriented generalist roles with specialized managerial and technical functions. Transportation remains a core domain; however, its professional expression is increasingly differentiated. At the same time, GIS and data analytics have become structural competencies rather than auxiliary skills. Collectively, these patterns indicate that contemporary planning practice increasingly requires the

Demonstration of a Recommendation Search Engine for Transportation Actions

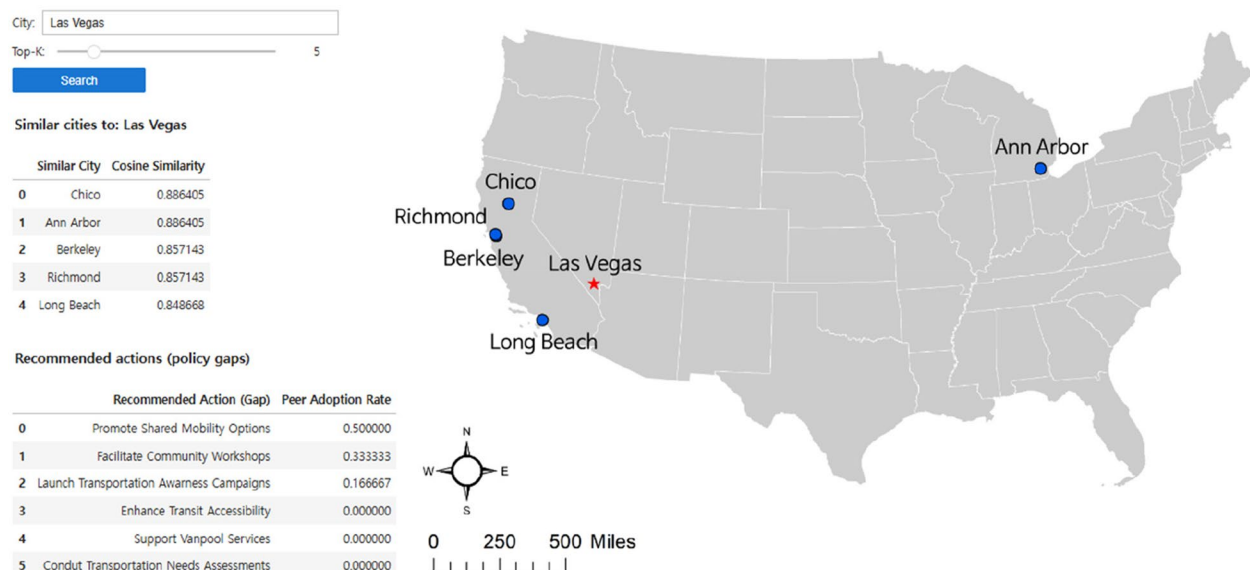


Fig. 7 Demonstration of a recommendation search engine for transportation actions

Demonstration of a Recommendation Search Engine for Energy Actions

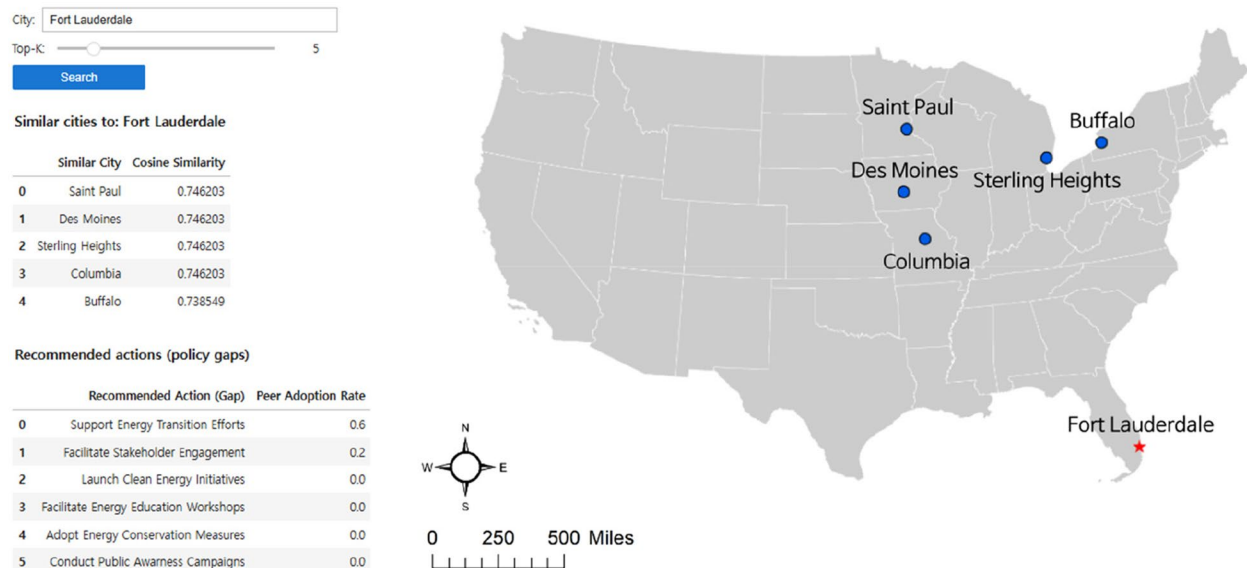


Fig. 8 Demonstration of a recommendation search engine for energy actions

integration of normative judgment with technical and regulatory expertise.

Despite rapid advances in GenAI, planning jobs continue to emphasize human-centered communication and judgment as core professional responsibilities. An examination of job descriptions shows that planners are commonly expected to perform communicative roles, including managing and evaluating strategies, supporting departmental functions, and distributing information while anticipating and correcting misinformation. Of the 83 job postings, 48 (58%) placed notable emphasis on communication-oriented responsibilities. The most frequently cited requirements included internal and external stakeholder communication, strong verbal and written communication skills, cross-functional coordination, and public-facing communication. Technical communication, particularly the ability to translate analyses and integrate data, also appeared as an important expectation. Such responsibilities align with Forester's (1982) view of planners as communicative actors, underscoring the enduring need for human-to-human interaction in planning practice. While recent advances in GenAI and LLMs are reshaping plan-making processes and affecting workforce composition (Zheng et al., 2025), positions that require human oversight, interpretation, and trust-building are likely to persist (Verma & Yadav, 2025). Similar to participatory planning traditions, emerging approaches to participatory algorithm design point toward new forms of algorithmic governance (Lee et al., 2019). The adoption of these tools depends on varying levels of trust, reliance, and aversion toward AI systems (Klingbeil et al., 2024). Consequently, domain-specific planning roles are

expected to continue, with emerging responsibilities for the responsible use and governance of AI becoming an integrated and defining component of planning practice. However, these responsibilities are likely to be carried out more effectively by more experienced professionals, which may raise barriers to entry for junior practitioners in the labor market (Brynjolfsson et al., 2025).

The contrast between the job-posting analysis and the AI-generated policy review directly necessitates the development of the RAPAF pipeline. The job market requires planners to balance intensive technical duties, such as GIS analysis and zoning administration, with broad communicative roles. Consequently, practitioners likely face a severe time constraint, making it practically impossible to manually review hundreds of multi-jurisdictional policy documents to identify innovative community advocacy strategies or policy gaps. The RAG-LLM excels precisely at identifying complex normative priorities, such as structural barriers for marginalized groups (e.g., LIDAC, BIPOC), within extensive texts. RAPAF is purposefully designed to bridge this operational gap. By utilizing a LangChain-native RAG pipeline to automate the semantic extraction of policy, strategy, and action items, RAPAF assumes the heavy analytical burden of large-scale document review. Ultimately, this framework resolves the tension identified in our empirical findings: it processes massive volumes of textual data so that planners can dedicate their limited cognitive bandwidth to the human-centric engagement, normative judgments, and strategic planning strictly demanded by their profession.

Second, this study identified common elements in climate equity or action plans across the U.S. and applied semantic analysis to examine their content. Topic modeling results based on tables of contents confirm that transportation and energy, particularly in relation to greenhouse gas emissions reduction, are consistently emphasized policy domains. These findings demonstrate the translation of established theoretical frameworks into planning practice (Damsø et al., 2016; Deetjen et al., 2018; Wu, 2023). Based on the semantic review and sentiment classification of direct quotations from the documents, policy texts predominantly employ affirmative language, with relatively few instances of negative semantics used to acknowledge limitations or challenges.

In practice, electric vehicle charging emerges as a recurring area of attention within transportation-related policy content. Despite political pushback against electric vehicles, their adoption continues to grow; however, this expansion also raises equity concerns related to disproportionate access to charging infrastructure and electric vehicle ownership (Nazari-Heris et al., 2022). Addressing these challenges is integral, and policymakers should consider leveraging overlooked action items such as public awareness campaigns, needs assessments, and impact studies. For energy-related action items, renewable energy is widely acknowledged as a priority; however, technical dimensions, such as grid modernization and efficiency improvements, are less frequently addressed, possibly due to the higher levels of specialized knowledge required in electrical engineering and grid systems. Based on this trend, ensuring that renewable energy systems are environmentally sustainable across their full life cycle is critical (Pehnt, 2006). Moreover, technical initiatives such as grid modernization encompass multiple interrelated elements, including infrastructure, operational processes, business models, and regulatory actions (Agüero et al., 2017). Addressing these challenges effectively requires engagement from multiple stakeholders to design more concrete and actionable policy interventions.

We further demonstrated that the use of LLMs fine-tuned with RAG frameworks can effectively support identification of community advocacy in equity planning. By implementing RAPAF and applying deterministic parameter settings, the approach improves accuracy and reduces hallucination risks commonly associated with direct LLM use (Zhao et al., 2024). Recent studies have demonstrated that equipping AI models with advanced reasoning capabilities to process unstructured and multimodal data can significantly augment evidence-based planning, ultimately driving the development of autonomous urban intelligence systems (Ki & Chen, 2026; Li et al., 2024, 2026). This capability is particularly important in the normative context of equity planning, where explicitly recognizing disadvantaged groups and their

spatial characteristics is essential for meaningful implementation (Brooks, 2002; Loh & Kim, 2021). When policy documents are embedded in a vector database (e.g., using LlamaIndex), the system enables planners to retrieve localized, context-specific information based on targeted queries, further mitigating hallucination risks compared to standalone LLM services. Similar to how human planners develop plan equity evaluation tools through systematic content analysis (Loh & Kim, 2021), this approach can be extended to perform comprehensive evaluations at significantly greater speed and scale.⁵ The use of affirmative versus negative language by virtual planners can be readily adjusted through the tuning of temperature parameters. However, prior to institutional deployment, it remains necessary to assess levels of trust in these systems through validation of accuracy (Deng et al., 2025) and the application of risk metrics tailored to AI governance contexts (Mohawesh et al., 2025; Shin, 2023).

Third, this study demonstrates the use of LLMs to develop a recommendation system for matching policy content across climate equity or action plans in different cities. The system allows users to adjust the number of recommended cities based on similarity, and it can be extended to evaluate policies, strategies, and action items separately. The results suggest that policy practices often exhibit geographically patterned similarities. For example, when examining similarities in transportation-related action items for Las Vegas, several cities in California, including Chico, Richmond, Berkeley, and Long Beach, were identified as comparable cases. Similarly, when Fort Lauderdale was used as a reference city for energy-related action items, a cluster of cities located primarily in the eastern U.S. emerged, including Saint Paul, Des Moines, Columbia, Sterling Heights, and Buffalo. The observed geographic clustering does not imply causality but highlights how regional contexts may influence the convergence of policy practices across jurisdictions. This pattern is consistent with the view that policymaking is mobile and shaped through policy conversations that circulate, with variation, across geographic contexts (Temenos & McCann, 2013).

It was also observed that the terminology and specificity of action items varied across cities. For example, in the case of Las Vegas (City of Las Vegas, 2026), explicit references to workshops were found only in Berkeley and Long Beach. Even between these two cities, the framing differed: Berkeley emphasized commission-hosted workshops, whereas Long Beach prioritized stakeholder workshops integrated with broader public outreach (City

⁵This approach requires supporting technical infrastructure, including server resources, vector databases, and data management systems for institutional deployment.

of Berkeley, 2009; Long Beach, 2026). More generally, all reviewed plans included some form of community engagement, though described differently, such as Chicago's "community engagement for sustainability planning," Ann Arbor's "public engagement throughout the climate action plan process," and Richmond's "community engagement around resilience and climate actions" (City of Ann Arbor, 2026; City of Chicago, 2026; City of Las Vegas, 2026; City of Richmond, 2026). Similar patterns emerged for mobility-related actions. Chicago emphasized improvements to bicycle networks, Long Beach highlighted the expansion of bikeway infrastructure, and cities such as Ann Arbor, Berkeley, and Richmond adopted broader terms such as "promoting public transit" or "enhancing multimodal transit connections," (City of Ann Arbor, 2026; City of Berkeley, 2009; City of Chicago, 2026; City of Richmond, 2026; Long Beach, 2026) which implicitly encompass cycling infrastructure improvements (Buehler & Hamre, 2015). These variations highlight an important implication: cities often pursue substantively similar climate or mobility goals but articulate them through locally specific language and institutional traditions.

The recommendation system developed in this study offers practical value as a policy learning and diagnostic tool. Rather than prescribing optimal policies, it enables planners to identify peer cities with similar policy profiles and examine which actions are present or absent in their plans. By surfacing overlooked or underutilized action items from comparable jurisdictions, the system can support reflective planning, peer benchmarking, and iterative policy refinement. The system is meant to help planners make decisions, not take their place. It provides an evidence-based starting point for exploring alternative policy actions while allowing planners to account for local context, institutional capacity, and political feasibility. By integrating agentic AI systems capable of analyzing building footprints and processing multimodal urban data (Li et al., 2024, 2026), it may become possible not only to generate planning recommendations but also to automatically evaluate urban morphology, built environments, and broader landscape characteristics.

The RAPAF-based recommendation engine is structured around three core pillars: interpretability, practical usability, and policy application. At its technical core, the system transcends traditional keyword-based metrics by leveraging RAG to evaluate the nuanced semantic context of explicit policy clauses. This interpretive depth directly addresses the persistent 'time-lag' in policy learning identified by Shipan and Volden (2012), which often delays the adoption of urban innovations. By identifying jurisdictions that have remained isolated from contemporary trends or exhibit low similarity to peer groups, the engine serves as a catalyst for policy diffusion,

systematically reducing the 'search friction' that typically hinders the cross-jurisdictional transfer of ideas.

Furthermore, the engine's ability to provide explicit source attributions empowers policymakers to foster evidence-based governance while significantly lowering administrative barriers. The practical value of such AI-integrated systems is supported by empirical evidence: LLM services have been shown to reduce professional documentation time by 40% while increasing quality by 18% (Noy & Zhang, 2023), with automated data extraction performing ten to a hundred times faster than manual encoding (Korini & Bizer, 2025). In actual practice, this system serves as a 'force multiplier' for policy benchmarking; for instance, an agency developing a Climate Equity Plan could use RAPAF to instantly identify how cities with similar demographic profiles have operationalized 'community heat resilience.' Instead of starting from a blank slate, planners can systematically review and synthesize the explicit policy clauses recommended by the system, ensuring their local plan is informed by the most relevant national best practices while focusing their expertise on contextualizing these narratives for local implementation.

However, successful implementation requires acknowledging the digital divide, as cities with robust digital infrastructures are currently better positioned to leverage these tools. While the system achieved an overall average accuracy of 70%, its performance varied significantly across semantic tiers; specifically, it demonstrated exceptional capability in exhaustive retrieval at the granular level, with a 94% recall rate for category-action classification. This performance gap suggests that while AI can effectively handle labor-intensive tasks, the precise selection of higher-level policies remains a nuanced challenge. Consequently, future implementations should prioritize a 'human-in-the-loop' framework. This collaborative environment would enable planners to audit AI-generated insights and determine the appropriate level of intervention, ensuring that technological efficiency does not compromise the rigor of professional planning judgment.

Finally, the scalability of RAPAF extends beyond large-scale urban systems in the U.S. and China. It offers profound utility for regions facing demographic and structural shifts, such as South Korea's declining birth rates or the shrinking suburban landscapes in parts of Japan. In these contexts, the system can provide critical governance support for compact city planning, ensuring that even smaller-scale jurisdictions can navigate complex urban transitions with data-driven precision.

6 Conclusion

Questions about how AI can assist policymaking gave rise to this study. We envision urban systems evolving toward greater intelligence, with existing tools and

infrastructure increasingly integrated with LLMs. Private companies have already begun adopting such technologies in their operations, which has intensified the need for orchestrating multiple AI agents and supporting infrastructure to ensure reliable performance, including next-generation communication networks such as 6G (Tarkoma et al., 2023). Planners primarily operate within the public sector and carry obligations to promote public goods (Brooks, 2002). By evaluating climate equity plans across the U.S. and analyzing planning-related job postings, this research validates the RAPAF framework as a practically viable tool and offers critical insights into the evolving roles of planners prior to conceptualizing the notion of 'virtual planners.'

This study contributes to the field by demonstrating how LLM-based RAG systems can support comparative learning, diagnostic evaluation, and policy recommendation in climate equity planning. By structuring policy documents into policy–strategy–action components and implementing a content-based recommendation system, the study moves beyond keyword-based assessments toward actionable policy insights. Importantly, the findings position AI not as a substitute for planners' normative judgment but as an infrastructure that augments cross-city learning while highlighting institutional, workforce, and governance considerations for public-sector adoption. Practically, the framework transforms the labor-intensive process of policy benchmarking into an efficient, source-attributed workflow, reducing 'search friction' and allowing planners to focus on high-level strategic synthesis. We further suggest directions for urban AI planners regarding the institutional adoption of AI-embedded systems in practice.

Nevertheless, several limitations should be noted. First, although the RAG pipeline was configured with deterministic parameters (temperature=0) to enhance reproducibility, responses may still be sensitive to design choices such as text chunking strategies and chunk size. This reflects a broader trade-off inherent in retrieval-based LLM systems between contextual granularity and stability. Second, the analysis relies on a specific LLM configuration at a given point. As LLMs continue to evolve, future model updates may influence semantic interpretations and retrieval behavior, raising questions of longitudinal reproducibility. Third, policy identification in this study is based on LLM-driven semantic interpretation rather than manual human coding. While this approach enables scalable analysis across large document collections, it may yield classifications that differ from those produced through traditional human auditing. This distinction highlights an important methodological tension between interpretive consistency and analytical scalability. While benchmarking against non-LLM baselines (e.g., LDA or TF-IDF) was considered, direct quantitative

comparison is impractical due to fundamental gaps in operational mechanics and output granularity. Unlike frequency-based models that yield isolated tokens, the proposed framework performs the contextual reasoning necessary to decode complex, hierarchical policy structures (Egger & Yu, 2022; Williams et al., 2024). Formally, the Bag of Words (BoW) representations underlying TF-IDF and LDA are invariant under token permutation, $BoW(\sigma(d)) = BoW(d)$, which collapses syntactically distinct expressions (e.g., the action "support electrification of buildings" versus the noun-phrase policy "building electrification support") into identical feature vectors. Finally, the demonstrated recommendation system adopts a content-based approach without incorporating explicit geographic or spatial information. While this design choice aligns with the study's focus on textual policy similarity, alternative approaches, such as collaborative filtering or spatially informed recommendation models, may surface different patterns of policy convergence. These limitations highlight important directions for future research.

Supplementary Information

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Supplementary Material 1.

Authors' contributions

The authors confirm contribution to the paper as follows: Seung Jun Choi: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. Sangyup Lee: Conceptualization, Validation, Writing – original draft, Writing – review & editing.

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Data availability

The data used in this study are available upon reasonable request.

Declarations

Competing interests

The authors declare no conflict of interest.

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